

StrengthWise

The AI strength coach that knows your sport,
remembers your history, and cites its sources.

YOUR DATA

+

TRAINING SCIENCE

=

CITED COACHING

Anar Shyraliiev April 2026 Grip Sport · Armwrestling · Powerlifting

Niche athletes train hard. Their data works against them.



Data trapped in spreadsheets

Years of training logs — sets, reps, RPE, notes — that never surface insights automatically.



Generic AI forgets everything

ChatGPT gives generic advice, knows nothing about gripper closes or armwrestling pronation, and forgets after every session.



Coaching knowledge stays siloed

Powerlifting periodization that works for grip peaking — no tool makes that cross-domain connection.



Logging friction kills consistency

Anything over 60 seconds per session gets abandoned. Most apps take 3–4 minutes.

980K+

COMPETITIVE
POWERLIFTERS
(OPENPOWERLIFTING DB)

\$150M

ARMWRESTLING
EQUIPMENT
MARKET (2025)

590K

REDDIT SUBSCRIBERS
R/POWERLIFTING +
R/ARMWRESTLING

\$4.4B

GLOBAL STRENGTH
SPORTS
MARKET (TAM)

0

tools with niche depth + personal memory + knowledge
RAG

Every competitor is missing at least two dimensions.

TOOL	NICHE SPORT DEPTH	PERSONAL MEMORY / RAG	KNOWLEDGE BASE RAG	FREE TIER
⚡ StrengthWise	♦ YES	♦ YES	♦ YES	♦ YES
ChatGPT / Claude	× No	× No	× No	✓ Yes
Juggernaut AI	~ Powerlifting only	× No	× No	× No
Whoop Coach	~ Partial	✓ Yes	× No	× No
Renaissance Periodization	~ Powerlifting only	× No	× No	× No
TrainHeroic	~ Partial	× No	× No	× No
Strong / Hevy	× No	× No	× No	✓ Yes
ArmProgress	✓ Armwrestling	× No	× No	✓ Yes
GripBoard Community	✓ Yes	× No	× No	✓ Yes
Fitbod / MyFitnessPal	× No	~ Partial	× No	✓ Yes

The gap: No tool occupies all four YES columns simultaneously. StrengthWise is the only one — by design.

Dual-source RAG coaching, built for athletes who train alone.

01 Dual-Source RAG

Every answer fuses your personal training history (DynamoDB sessions) with a curated strength science knowledge base (FAISS). Both sources are cited in the response.

02 Persistent Memory That Compounds

Your data accumulates across sessions. Confidence level improves with every log: "Based on 3 sessions..." → "Based on 60 sessions, a clear pattern shows..."

03 Niche Sport Depth

Knows gripper closes, hub lifts, AW pronation/supination, powerlifting peaking cycles. Cross-domain transfer: PLing periodization → grip programming.

"Why did my grip close stall in the last cycle?"

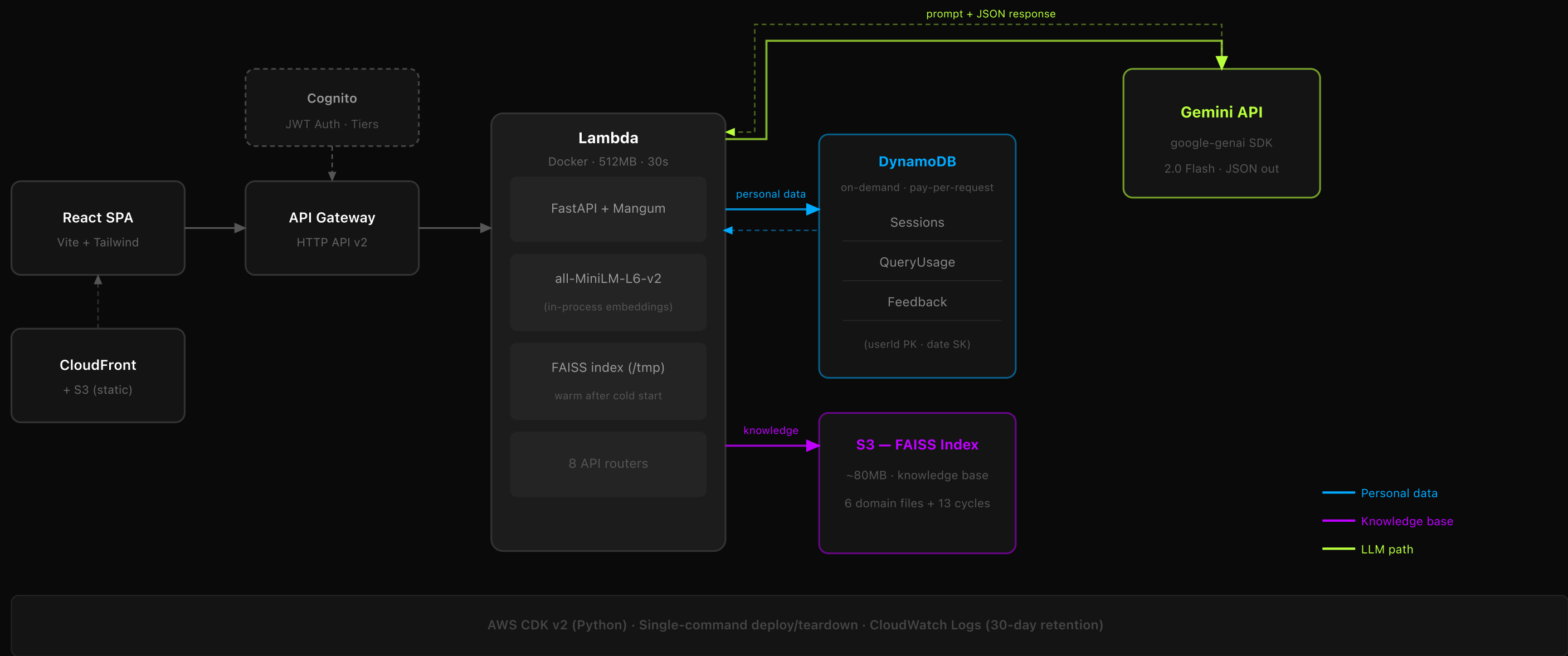
YOUR DATA

Your last 3 sessions show consistent RPE 9–10 on gripper closes with no volume reduction — classic accumulation without deload. Your session on Mar 14 hit 8×3 at 10 RPE.

TRAINING SCIENCE

Grip-specific fatigue accumulates faster than lower-body strength (Zatsiorsky, 2006). A 5–7 day deload at 40% volume is recommended after 3+ weeks of high-RPE grip work.

Serverless by design. Zero idle cost.



One query. Two sources. One cited answer.

①

User Query

Athlete types question
→ sanitized → POST
/query
Rate limit checked
against QueryUsage
table

②

Embed + FAISS Search

all-MiniLM-L6-v2
encodes query →
cosine similarity →
top-3 knowledge
chunks from FAISS
index

③

Personal Context

DynamoDB → last 10
sessions for userId →
structured as training
history text

④

LLM Synthesis

Both sources injected
into structured prompt
→ Gemini 2.0 Flash →
JSON with response +
citation arrays

⑤

Cited Response

■ YOUR DATA pills +
■ TRAINING
SCIENCE pills
rendered inline

```
{ "response": "Your grip stall is classic overreaching without deload...", "confidence": "medium" // <= 30 sessions logged "personal_citations": [{"sessionId":"2026-03-14","exercise":"gripper close","sets":8,"rpe":10}], "knowledge_citations": [{"source":"grip_programming.md","principle":"high-RPE accumulation deload protocol"}] }
```

612 indexed chunks. 40 documents. 37 exercises. Built for niche sports.

612

KNOWLEDGE CHUNKS
(FAISS INDEX)

40

SOURCE DOCUMENTS
IN CORPUS

37

SPORT-SPECIFIC
EXERCISES

19

BACKEND TEST
MODULES

KNOWLEDGE CORPUS BREAKDOWN

CATEGORY	DOCS	CHUNKS
Core Strength Science	6	96
Training Systems	8	154
Historical Cycles	13	162
Niche Sport Programs	2	39
Sport Programming	2	34
Research & Reference	5	79

EXERCISE LIBRARY — 4 SPORTS

GRIP · 8

ARMWRESTLING · 9

POWERLIFTING · 10

GENERAL · 10

Embedding specs

model: all-MiniLM-L6-v2

dims: 384

index: FAISS IndexFlatIP (cosine)

size: << 1 GB (currently ~1 MB) — fits in Lambda /tmp

Demo Seed Dataset 77 sessions · 3 dev users: free (5 grip), onboarding (12 grip + AW), premium (60 — grip RPE stall → deload → rebound + full PL peaking cycle). Each user triggers a different AI confidence tier.

Four evaluator types. Three RAG strategies. Applied to StrengthWise.

EVALUATOR TYPES

TYPE	STRENGTHWISE IMPLEMENTATION
 Human	Thumbs up/down per query → /feedback endpoint. Target ≥ 95% positive rate
 Heuristic	Citation coverage rule: 100% of 3+ session users must receive ≥1 personal cite. Rule-based confidence tier assignment.
 LLM-as-Judge	Gemini evaluates answer faithfulness to retrieved chunks & helpfulness via structured prompt rubric. Applied to 50 reference Q&A pairs.
 Pairwise	Ablation study: Dual-RAG vs Personal-only vs Knowledge-only vs Vanilla LLM — same question, four conditions.

RAG EVALUATION USE CASES

Reference-free = no gold answer required; judge checks internal consistency

USE CASE	REF-FREE?	STRATEGY
Document relevance	✓	Online
Answer faithfulness	✓	Online
Answer helpfulness	✓	Online
Answer correctness	✗	Offline (50 Q&A)
Chain comparison	✓	Pairwise (ablation)

Confidence Calibration — Heuristic Rule

Low (1–5 sessions): "early indicators suggest..."

Medium (6–30 sessions): "looking at your recent data..."

High (30+ sessions): "a clear pattern shows..."

Freemium with metered AI coaching. Zero cost for the core habit.

FREE FOREVER

\$0

- ✓ Unlimited session logging
- ✓ 3 AI queries / day
- ✓ Full dual-source RAG quality
- ✓ Sport-native exercise library
- Export to CSV / PDF
- Unlimited queries

Onboarding: 10 queries / day for first 7 days

PREMIUM

\$7–10 / month

- ✓ Everything in Free
- ✓ Unlimited AI queries
- ✓ CSV + PDF export (roadmap)
- ✓ Priority response queue

Upgrade = one Cognito group change.
Rate limiting already implemented.

TEAM / COACH

ROADMAP

TBD

- ✓ Multi-athlete management
- ✓ Coach dashboard & analytics
- ✓ WAF/IFA integration
- ✓ White-label API

Phase 2 — B2B pivot after community traction.

Strategic insight: Logging is \$0 LLM cost — it builds the habit and personal data moat. AI queries are the monetized layer. A serious competitive athlete spending \$7/month on coaching is well below the pain threshold.

Break-even at 12 paying users.
Profitable at 25.

12-MONTH CONSERVATIVE MODEL

500

TARGET ACTIVE USERS (90-DAY)

5%

FREEMIUM → PAID CONVERSION

\$7.50

ARPU (MIDPOINT)

25

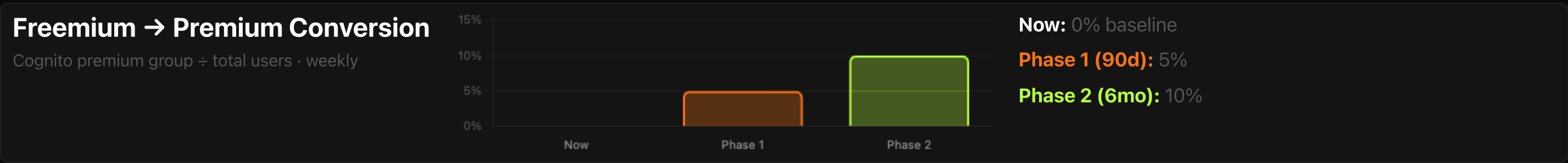
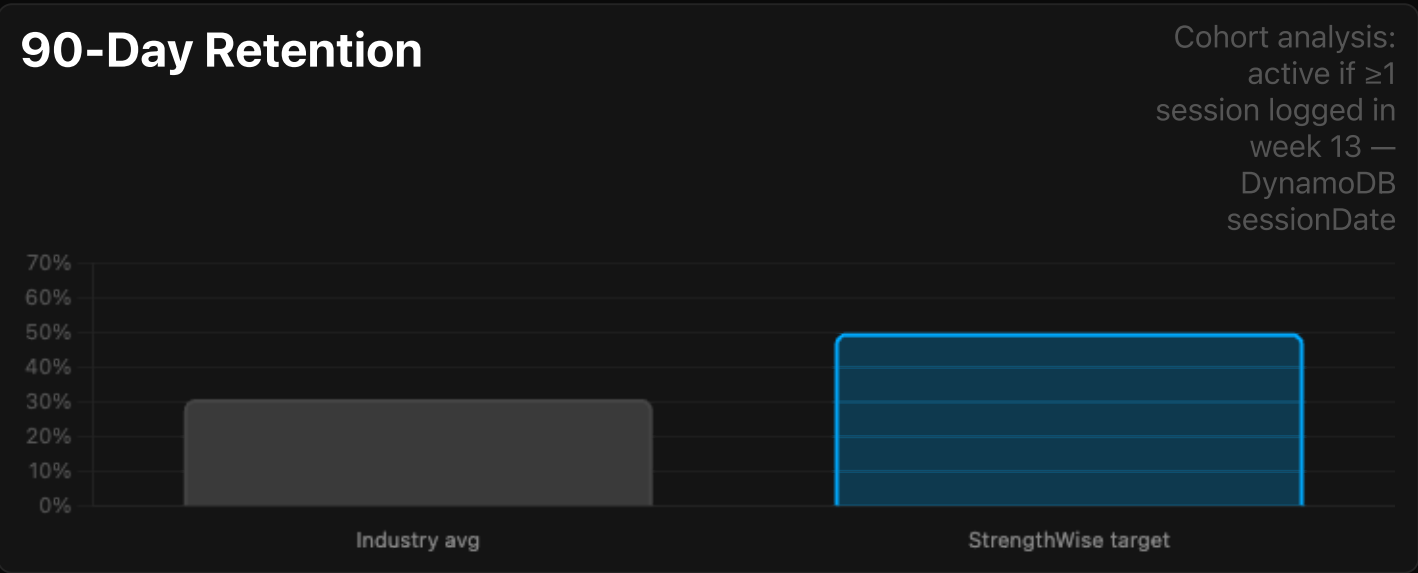
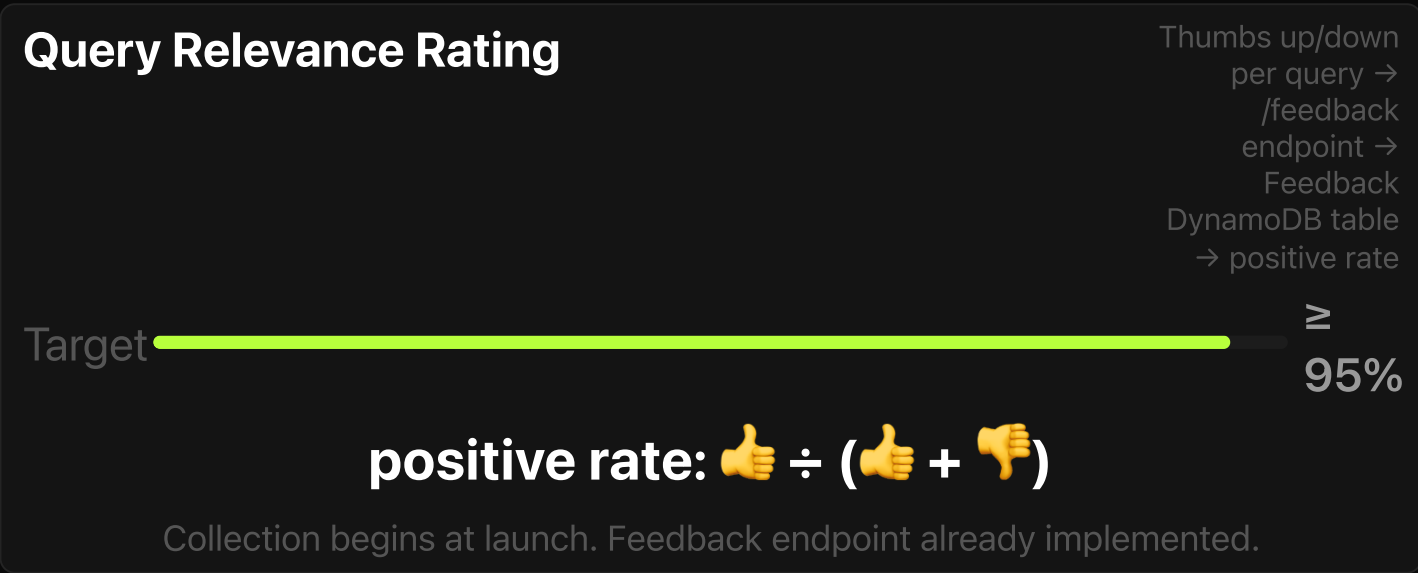
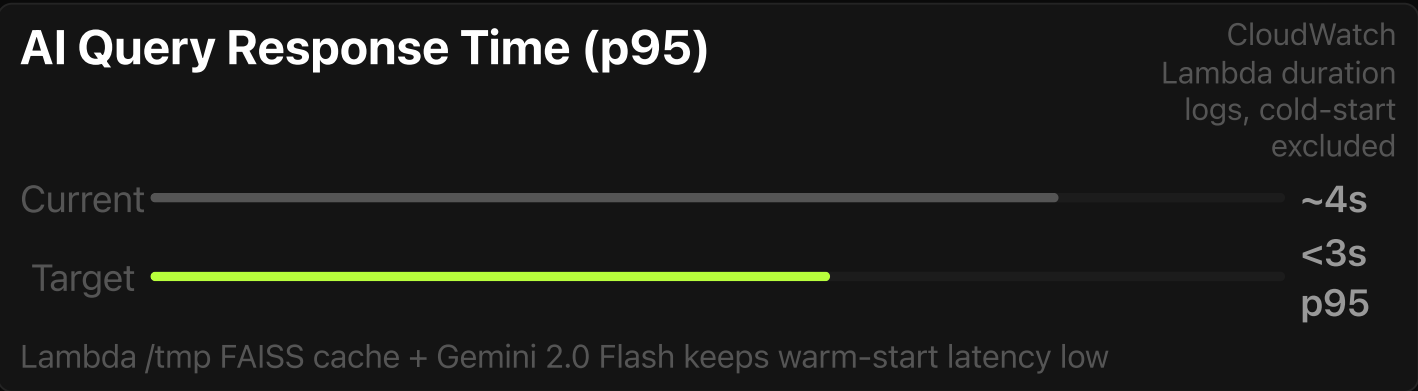
PAYING SUBSCRIBERS

MONTHLY COST STACK

Lambda (pay-per-request)	~\$5
DynamoDB (on-demand)	~\$8
Gemini API (main variable cost)	~\$40
CloudFront + S3	~\$12
Cognito (MAU)	~\$5
Miscellaneous	~\$20
Total	~\$90/mo

Break-even: 12 paid users ($\$90 \div \7.50). With 25 users: **\$187.50 revenue vs \$90 costs = 54% margin.**
At 5,000 users with the same architecture: margin approaches **80%** — Gemini API is the only meaningful variable cost. No re-architecture needed.

5 KPIs. Each with a measurement methodology.



Community-first growth.

Architecture that scales without re-engineering.

PHASE 1 — 3 TO 6 MONTHS

- M1 • Public Launch**
Deploy to AWS. Seed with 13 historical cycles as demo content. Post to r/GripTraining, r/armwrestling, GripBoard.
- M2 • Community Feedback Loop**
100 active users milestone. Weekly feedback sessions with grip and AW competitors. Iterate on exercise library.
- M3 • Structured Onboarding**
Sport selector → auto programming recommendation. Reduce time-to-aha-moment from session 3 to session 1.
- M4–6 • Analytics + Export**
Performance dashboard, CSV/PDF export, 500 users milestone, Stripe integration for premium.

PHASE 2 — 1 TO 2 YEARS

- Y1 • v2 Feature Layer**
Voice logging, XLSX bulk import, automated program generation, HRV/wearable integration, proactive pattern alerts.
- Y1 • B2B Pivot**
Coach dashboard, multi-athlete management, WAF/IFA governing body partnerships, white-label API for gyms.
- Y2 • Platform Expansion**
Multi-region DynamoDB, multi-model support (fine-tuned open models), competition tracking, community features.

500

users → profitable
12 paid = break-even

5K

users → hire 1 FTE
invest in wearables

50K

users → B2B pivot
institutional licensing

LIVE DEMONSTRATION

Let's see it live.

"Ask it something your coach never could."

" Why did my grip stall in the last cycle? "

" Design a 4-week AW program from my history "

" Compare my RPE trends to periodization science "

Questions?

- Deployed on AWS — CloudFront, Lambda (Docker), DynamoDB, Cognito, S3
- Zero fixed cost — every component is pay-per-request
- The only tool with niche sport depth + personal memory + knowledge RAG
- Break-even at 12 paying users (\$90 / \$7.50 ARPU)
- 5 measurable KPIs with existing instrumentation in the codebase

LIKELY QUESTIONS

Why Gemini 2.0 Flash? Strong reasoning at Flash pricing (~\$0.0004/query); Pro models are 15–20× more expensive

FAISS vs Pinecone? FAISS in Lambda /tmp = zero per-query vector DB cost

Data privacy? userId = Cognito UUID — no PII in training logs

Cold start? FAISS loads ~80MB from S3 on cold start, stays in /tmp after

📁 **gen-ai-project-set-university** · Anar Shyraliiev · 2026